Experimental Engineering Lab
Designing and Deploying Modular WebAR Experiences

By Darwin Lemus

Project Motivation and Goal

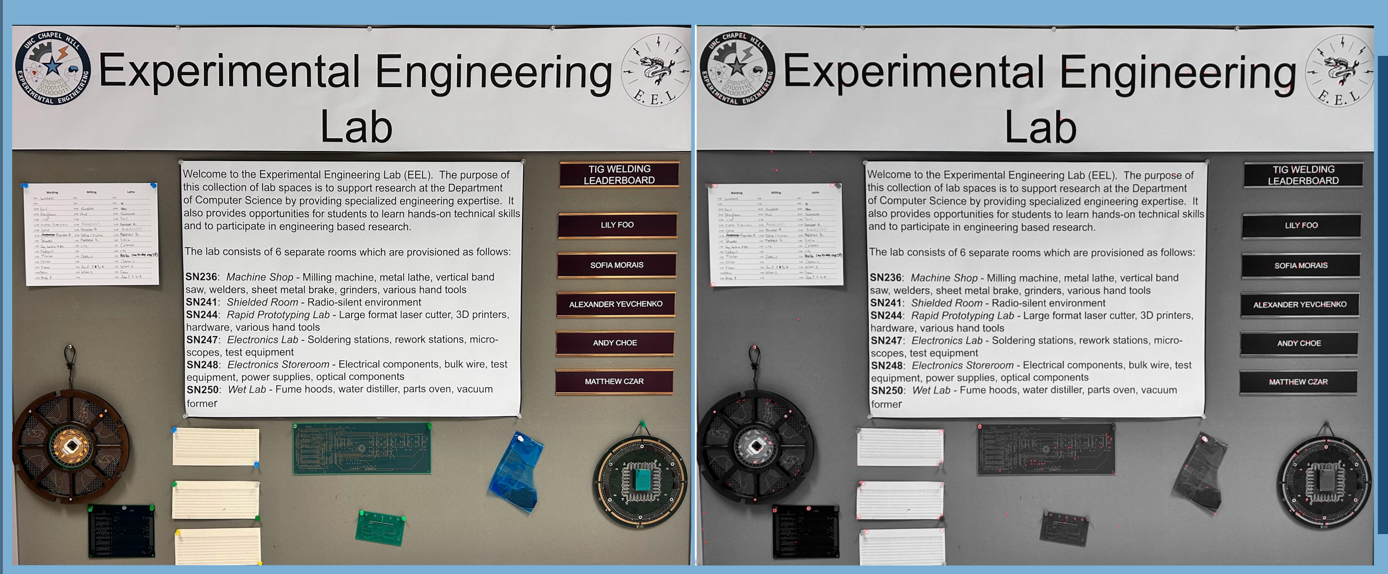
This project focused on building a lightweight, browser-based WebAR experience that could be easily deployed and reused in different spaces. Rather than relying on AprilTags or other fiducial markers, the system used natural visual targets already present in the room to create a less intrusive and more seamless AR experience. What began as a lower lobby AR installation developed into a modular framework for location-based WebAR.

Research and Technology Selection

- Compared multiple WebAR approaches through paper review and technical exploration
- Evaluated browser-based AR against downloadable software solutions
- Selected **A-Frame** for browser-based scene creation
- Selected **MindAR** for image-target tracking
- Chosen stack balanced **accessibility, lightweight deployment, and modularity**, which later supported development of a reusable AR poster toolkit

System Design and Implementation

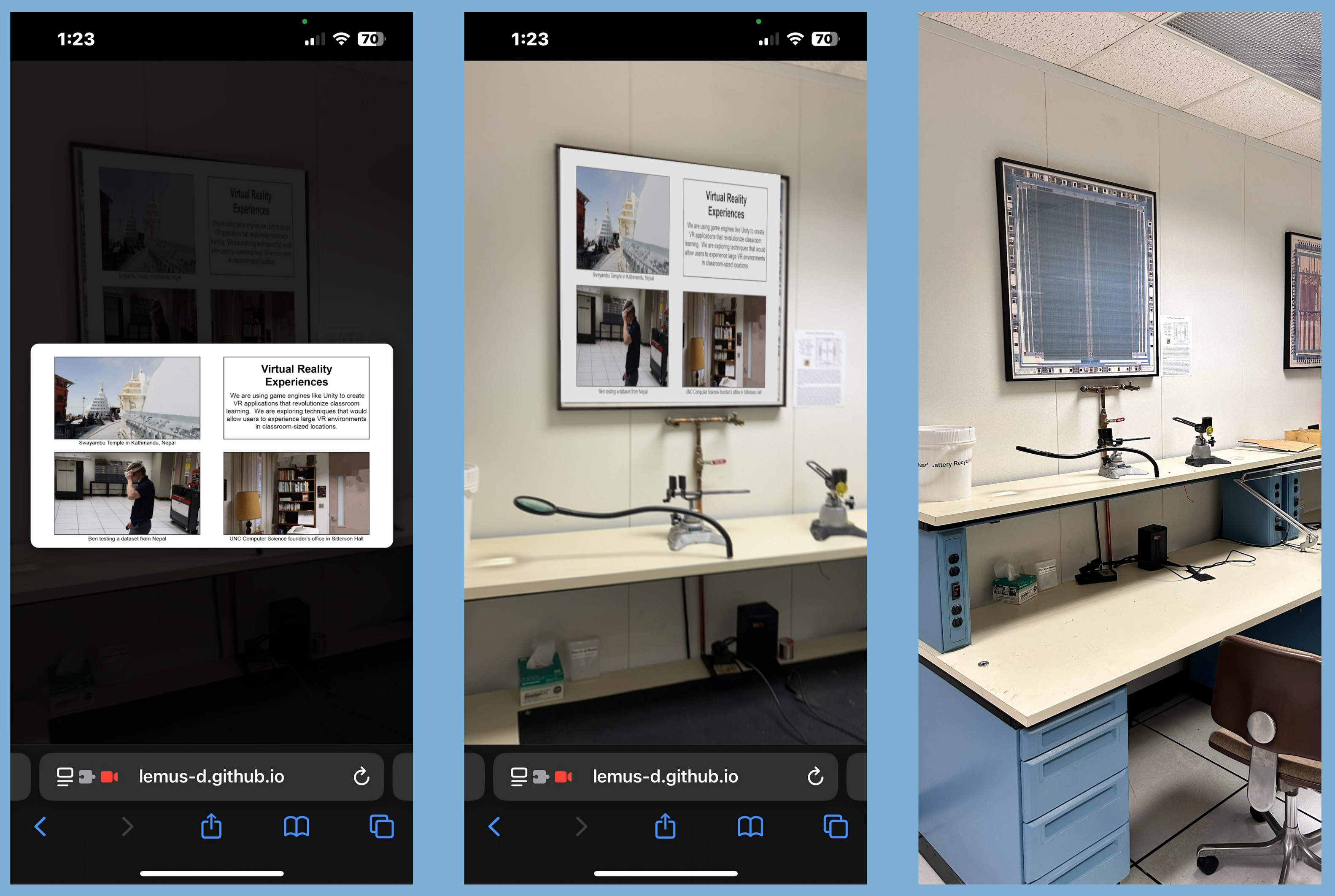
- Physical image targets trigger digital AR content in a mobile browser
- Supported media include **images, videos, and 3D models**
- Built using **A-Frame** for scene creation and **MindAR** for image-target tracking
- Includes an **admin dashboard** that lets users upload assets, pair media to targets, and adjust transforms
- Uses `config.json` to programmatically generate the viewer, so end users do not edit HTML directly
- Hosted on **GitHub Pages** for lightweight deployment and sharing



Original Image VS Compiled Target Image

Challenges, Testing, and Optimization

One of the biggest development challenges was reliable target tracking. Some images lacked enough contrast points for stable recognition, so selecting effective targets became an important design consideration. To improve stability and ease of use, I had people in the lab test the system throughout development and used that feedback to refine performance. I also tuned MindAR parameters such as `filterMinCF`, `filterBeta`, and `warmupTolerance` to reduce jitter and improve responsiveness.



Example of a digital poster in our lab

Scan Me!

Current Deployment and Future Work

The project is currently deployed in the EEL lab and SN020 using different targets and media. It has since been expanded into a reusable **AR poster toolkit** that users can clone, customize, and host on GitHub Pages. The toolkit includes a browser-based dashboard for pairing targets with images, videos, or 3D models and generating the final configuration without manually editing the main viewer code. Future work will focus on improving usability, expanding interaction options, and making the toolkit easier for non-technical users to adopt.